Design Manifesto in the Project

MY DESIGN POSITION

Symbiotic Design

“True user advocacy is not about ignoring business viability, but creating a mutually beneficial balance where company goals and user utility reinforce each other.”

Tower of Arithmetic is built on this idea: the game only succeeds if learners genuinely get better at maths and enjoy it. The learning is the game, not a toll-gate in front of it, so learner value and product success grow together.

View my visual manifesto (Figma) ↗

HOW THE BALANCE WORKS IN THIS GAME

Learners get

Real, curriculum-matched practice that builds confidence instead of anxiety.

The product gets

Players who return because progress feels rewarding, not because they are trapped.

Parents & teachers get

Honest progress, safe data and no pressure to pay.

The product gets

The trust schools and families need before they adopt a learning app.

Manifesto Principles 1–4

01 User Value Drives Success

Solve a real problem; success comes from trust, not exploitation.

IN TOWER OF ARITHMETIC

- Every spell is cast by solving a sum: maths is the core mechanic, not a pop-up quiz.
- Sums follow the CAPS curriculum for each grade, from Grade 1 to 11.

02 Build With Users

People affected by a product should shape it.

IN TOWER OF ARITHMETIC

- Learner and parent surveys drove the problem frame and personas.
- Usability tests with learners, plus an in-app beta notice that invites feedback.

03 Universal Accessibility

Aim for WCAG AAA from the start, not as a patch.

IN TOWER OF ARITHMETIC

- AAA-contrast palette, 44 px touch targets and screen-reader labels.
- Text size, read aloud, speech-to-text, handwriting input and a dyscalculia mode.

04 Research With Integrity

Honest, open research, never cherry-picked.

IN TOWER OF ARITHMETIC

- Open, non-leading survey questions; negative findings are reported too.
- Competitor gaps (e.g. punitive scoring) came from evidence, not assumptions.

Manifesto Principles 5–7

05 Never Stay Silent

Speak up when a design hurts users, and bring an alternative.

IN TOWER OF ARITHMETIC

- Instead of slashing a score after a mistake, a wrong answer gives a free hint from Orrin.
- “Show me how” opens a short lesson right where the learner got stuck.
- Puzzle doors cost no hearts: learners can try again calmly.

06 Fair Business, No Manipulation

Support fair income; never trap or trick users.

IN TOWER OF ARITHMETIC

- No pay-to-win, loot boxes, energy timers or streak guilt.
- Coins, items and potions are only earned by playing.

07 Transparent Data

Collect only what is needed and explain it plainly.

IN TOWER OF ARITHMETIC

- An account only needs an email, a username and a grade.
- The AI only sees the sum and the answer, never a name or email.
- API keys stay on the server, and nothing is pre-ticked.

Manifesto in Action

Three real trade-offs in Tower of Arithmetic, and how the manifesto settled them.

PRINCIPLES 1 & 5

Challenge vs. maths anxiety

THE TENSION

A game needs risk, but losing hearts for every wrong answer can raise maths anxiety.

MY DECISION

Fights keep hearts, but each mistake brings a free hint and a lesson link; puzzles never cost hearts.

THE TRADE-OFF

Slightly less tension, but learners stay and keep practising.

PRINCIPLE 3

Dark fantasy vs. readable text

THE TENSION

A moody, dark art style can easily mean low-contrast text.

MY DECISION

An obsidian background with gold and off-white text at AAA contrast, and shades behind text on artwork.

THE TRADE-OFF

Fewer subtle colours, but every learner can read every screen.

PRINCIPLE 7

Smart AI vs. privacy

THE TENSION

Personalised AI hints work better with more data about the learner.

MY DECISION

The AI gets only the sum, the answer and the grade; the summary uses topic results, never personal details.

THE TRADE-OFF

Less personal hints, but no child data leaves the game.